



DRILLING



Example 1

Blind Hole: $\varnothing 15 \times 45$ mm on a stainless steel part

Data:

$$k_c = 2800 \text{ MPa}$$

$$\varepsilon = 140^\circ$$

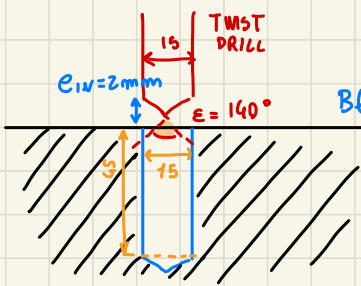
$$f = 0,2 \text{ mm/rev}$$

$$v_c = 60 \text{ m/min}$$

Determine:

- a) Chip thickness
- b) Force, torque and power
- c) Drilling time ($e_{in} = 2 \text{ mm}$)

EXTRA PATH IN



$\epsilon = 140^\circ$ tool tip angle

Blind hole to make $\rightarrow$ done by a Twist drill

hole depth of 45 mm

$$K_c = 2800 \text{ MPa}$$

$$f = 0,2 \text{ mm/REV}$$

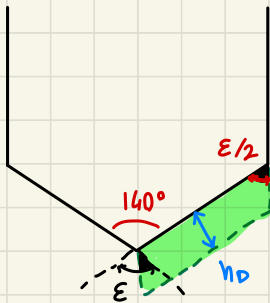
$$V_c = 60 \text{ m/min}$$

with diameter equal to the hole diam of 15 mm

extra path in

of $e_{1/2} = 2 \text{ mm}$ for drilling

a) CHIP THICKNESS



h_D measured $\perp$ to cutting edges

area of chip before deform.

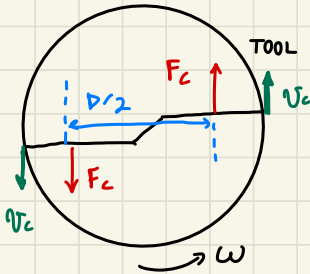
$$f_z = f/2 = 0,2/2 = 0,1 \text{ mm/REV}$$

2 cutting edges $Z=2$

$$h_D = f_z \cdot \sin(\epsilon/2) = 0,1 \cdot \sin\left(\frac{140}{2}\right) = 0,094 \text{ mm}$$

(usefull if we have to compute K_c)

b) from BOTTOM



couple of cutting forces F_c

with DISTANCE $D/2$

$$T_c = F_c \frac{D/2}{1000}$$

(torque given by the couple of forces! considering all the ARM of $D/2$)
in between

$$P_c = T_c \cdot \omega$$

$$F_c = K_c \cdot \underbrace{A_D}_{f_z \cdot D/2} = 2800 \cdot 0,1 \cdot \frac{15}{2} = 2100 \text{ N}$$

$$T_c = 2100 \frac{15/2}{1000} = 15,75 \text{ N.m}$$

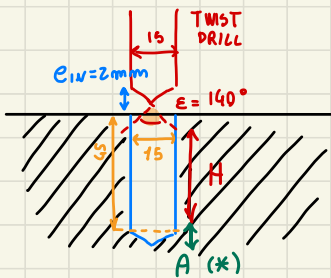
from V_c information, we can compute $n = \frac{1000 \cdot 60}{\pi \cdot 15} = 1273 \text{ rpm}$

$$\Rightarrow \omega = \frac{2\pi \cdot n}{60} = \frac{2\pi}{60} \cdot 1273 = 133,33 \text{ rad/s}$$

$\uparrow$
peripheral external diameter

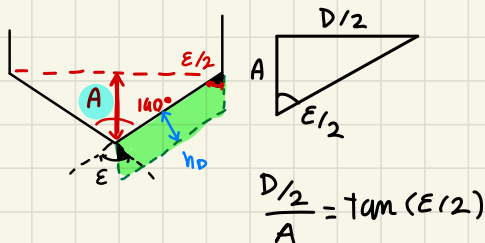
$$P_c = T_c \cdot \omega = 15,75 \cdot 133,33 = 2099,95 \text{ W} \approx 2,1 \text{ kW (real approx)}$$

Drilling Time



for overall process we start from
 Ein distance, move along feed for 45mm
 than move more of A (*)

Can be computed from the drawing



$$\Rightarrow A = \frac{D/2}{\tan(E/2)} = \frac{15/2}{\tan(140/2)} = 2,73 \text{ mm}$$

$$t_m = \frac{e_{in} + H + A}{f} = \frac{2 + 45 + 2,73}{0,2 \cdot 1273} \cdot 60 = 11,7 \text{ s}$$

total
feed of
the tool

making of the
HOLE time...

BUT we have to come BACK and
 retrieve the tool from the HOLE

if keeping

same condition $t_{m_{tot}} = 2 t_m$

or you can

come back moving the tool with a Rapid motion! V_R

$$\text{RETRIVAL TIME } t_R = \frac{2 + 45 + 2,73}{V_R} \text{ same length}$$

rapid speed motion along axis

OR IF same condition when retrieval $t = 2 t_m$



Example 2

(simple situation with also)

HOLE finishing

right operation for finishing, remove more material, defining starting point of finish

we need also FINISH!

range of hole dimension

(usually IT10 quality when hole making)

Tolerance quality!
(variability limit)

Through hole drilling and reaming (H7):

$\varnothing 20 \times 25$ mm on carbon steel plate (HB200)

we START reasoning on FINISHING (mach. allowance) and decide the tool to use for hole creation

Drilling:

$$D = 19 \text{ mm}$$

$$k_c = 2100 \text{ Mpa}$$

$$\varepsilon = 140^\circ$$

$$f = 0,1 \text{ mm/rev}$$

$$v_c = 120 \text{ m/min}$$

Reaming:

$$D = 20 \text{ mm}$$

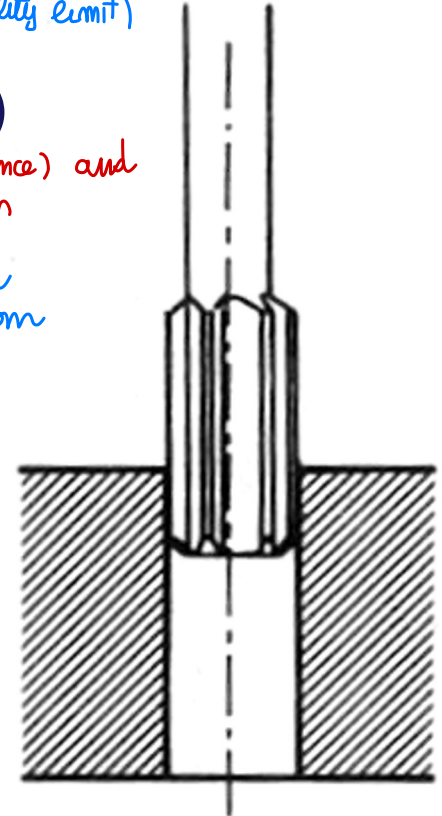
$$k_c = 2100 \text{ Mpa}$$

$$Z = 20$$

$$f = 1,5 \text{ mm/rev}$$

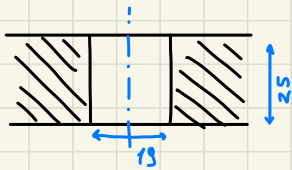
$$v_c = 180 \text{ m/min}$$

to reach final dimension



Determine the cutting force, torque, and power for the two machining operations.

• DRILLING Through hole (Thickness equal to hole height (25 mm))



to have a final $\phi 20$ Diam, during drilling we remove until $\phi 19$ remaining 0.5 mm of radius allowance (material allow on cylindr surface)

• F_c, T_c, P_c ?

$$K_c = 2100 \text{ MPa}$$

$$\varepsilon = 140^\circ$$

$$f = 0.1 \text{ mm/REV}$$

$$V_c = 120 \text{ m/min}$$

$$A_D = \frac{D \cdot f}{4} = \frac{19 \cdot 0.1}{4} = 0.475 \text{ mm}^2$$

$$F_c = K_c A_D = 2100 \cdot 0.475 = 997.5 \text{ N}$$

$$T_c = F_c \frac{D/2}{1000} = 997.5 \frac{19/2}{1000} = 9.5 \text{ N m}$$

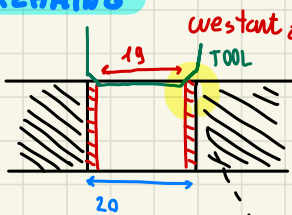
$$P_c = T_c \cdot \omega$$

$$\omega = \frac{1000 \cdot 120}{\pi \cdot 19} = 2010 \text{ rpm} \rightarrow \omega = \frac{2\pi}{60} \cdot 2010 = 210.5 \text{ rad/s}$$

$$P_c = 9.5 \cdot 210.5 = 1999.75 \text{ W} = 2 \text{ kW}$$

2ND OPERATION

REAMING



we start from the hole with $\phi 19$ mm of DRILLING!

$$K_c = 2100 \text{ MPa}$$

$$f = 1.5 \text{ mm/REV}$$

$$V_c = 180 \text{ m/min}$$

$$Z = 20 \text{ (cutting edges of the TOOL)}$$

$$D = 20 \text{ mm}$$

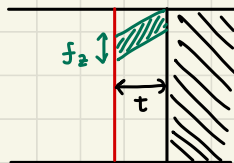
lot of teeth of tool

on this situation, the CHIP AREA

IF we zoom on the material to remove

Tool to have full def we should know ε (usually here $\varepsilon \approx 180^\circ$)

$$A_D = \left(\frac{f_z}{20} \right) \cdot \left(\frac{20 - 19}{2} \right) = 0.0375 \text{ mm}^2$$



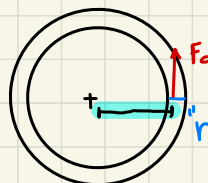
$$t = 0.5 \text{ mm}$$

$$= \frac{D_e - D_i}{2} = \frac{20 - 19}{2} = 0.5 \text{ mm}$$

$$f_z = f/Z \rightarrow A_D = f_z \cdot t$$

$$F_c = K_c \cdot A_D = 2100 \cdot 0.0375 = 78.75 \text{ N (small value \& cutting edge)}$$

to compute TORQUE we should keep in mind that we are cutting in ring



COMPUTE

contribution of single cutting force F_c to all T_c

$$r \text{ "average radius": } r = \frac{D_e + D_i}{4} = \frac{20 + 19}{4}$$

all 2 cutting edges works simultaneously

$$T_c = 20 \cdot \frac{78,75 \left(\frac{20+19}{4} \right)}{1000} = Z \cdot \frac{F_c \left(\frac{D_e + D_i}{4} \right)}{1000} = 15,4 \text{ N}\cdot\text{m}$$

Different

working condition than Drilling...

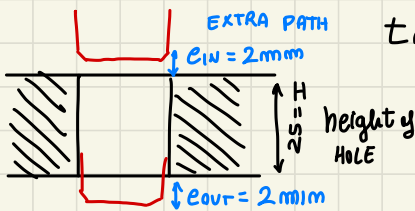
$$n_R = \frac{1000 \cdot 180}{\pi \cdot 20} = 2865 \text{ rpm}$$

$$\omega_R = \frac{2\pi}{60} \cdot n_R = \frac{2\pi}{60} \cdot 2865 = 300 \frac{\text{rad}}{\text{s}}$$

so once found motion value...

$$P_{c_R} = T_c \cdot \omega_R = 15,4 \cdot 300 = 4620 \text{ W} = 4,62 \text{ kW}$$

IF in through hole we want discuss machining time



$$t_m = \frac{e_{in} + e_{out} + 2S}{f \cdot n} = \frac{2 + 2 + 2 \cdot 5}{1,5 \cdot 2865} \cdot 60 = 0,4 \text{ s}$$

{ FAST OPERATION }

- classical hole making situation...

↑ classical enlarging

REAMING (finishing)

@ FIRST approx also

when doing threading

we can consider it at the same way!

MILLING



Example 3

feature called → "STEP" when creating 2 ORTHOGONAL surfaces on one edge of the workpiece

The groove shown in figure should be made using a milling operation. Determine the corresponding cutting power.

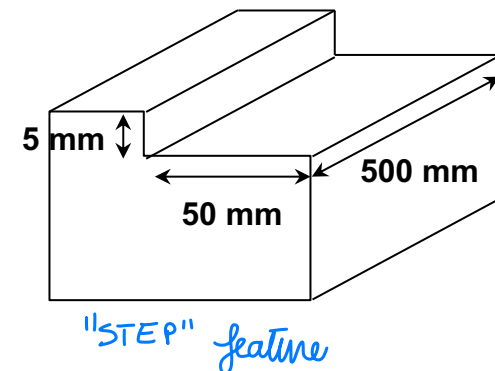
DATA:

Tool data:

- Number of tooth $Z = 5$;
- Entering angle $\kappa_r = 90^\circ$;
- Mill diameter $D = 60$ mm;

Machining parameters:

- Cutting pressure $k_{c0,4} = 1600$ MPa
- Coefficient $x = 0,29$
- Feed per tooth $f_z = 0,1$ mm/rev

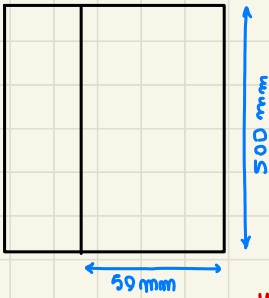


Maximum chip thickness [mm]	0,05	0,1	0,2
Cutting speed [m/min]	305	275	225

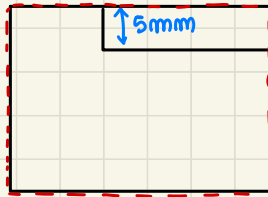
Determine the cutting force, torque, power, and time for this machining operation.

We wanna create that step object

TOP VIEW



SIDE VIEW



starting point is a parallelepiped block!
WE are making a STEP on the edge
WORKING external edge



We are performing a Face Milling operation

(creating both surfaces)

prevalently face milling

BUT also slab milling

Respect vertical plane

$$Z = 5$$

$$D = 60 \text{ mm} \quad K_r = 90^\circ$$

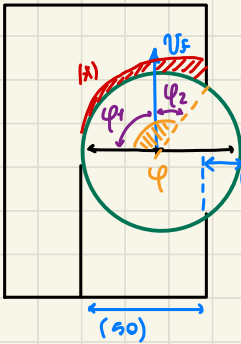
Must be 90° to create the straight right angle between the surfaces

$$K_{c0.4} = 1600 \text{ MPa} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{KRONENBERG}$$

$$\chi = 0.29$$

$$f_z = 0,1 \text{ mm/rev}$$

Looking the operation from the TOP:



(FACE MILLING situation)

tool ω
60
(10) portion outside of material

With an engaging angle φ that can be found...

moving vertically (feeding) feed motion

$$\varphi = \varphi_1 + \varphi_2$$

TOOL ROTATION? in this situation...

clock OR counter-clock wise?

WE want a counter clock wise Rotation, impact

← considering the material to remove (*)

IF CLOCK WISE

direction $n_0(\theta) = \theta$

at engagement point you have θ

thickness φ

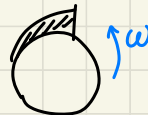
FRICION HEATING and so on...



so we prefer opposite rot. direction

(avoiding the issue

about the forces and possible GAP over transmission)

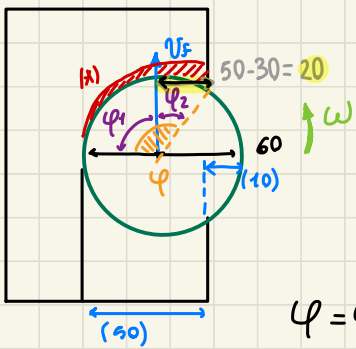


BETTER!

NOT always

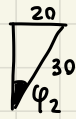
possible, IF you can choose, IT is better counter clock wise!

$$\left\{ \begin{array}{l} \varphi_1 = 90^\circ \\ \varphi_2 = ? \end{array} \right.$$



$$\varphi_1 = 90^\circ, \varphi_2 ?$$

$$20 = 30 \cdot \sin(\varphi_2)$$



$$\varphi_2 = \arcsin(20/30) = 0,63 \text{ rad} = 41,8^\circ$$

[rad] (we will use rad)

$$\varphi = \varphi_1 + \varphi_2 = 90 + 41,8 = 131,8^\circ = 2,3 \text{ rad}$$

$$\varphi_0 = \frac{2\pi}{Z} = \frac{2\pi}{5} = 1,25 \text{ rad} = 72^\circ \quad (\text{distance between each tooth})$$

$$z = \varphi / \varphi_0 = 131,8 / 72 = 1,84$$

$$1 < z < 2$$

OK WORK
Condition!

evaluate F_c, T_c, P_c, t_m ?

WORST CASE SCENARIO? (for $z \in (1,2)$)

We Refer to the fact that being $z=1,84$: One cutting edges is almost always working for most of the time 2 cutting edges simultaneously WORKING, position this on WORST case...

When $\begin{cases} \vartheta_1 = \varphi_0 / 2 = 72 / 2 = 36^\circ \\ \vartheta_2 = -\varphi_0 / 2 = -72 / 2 = -36^\circ \end{cases}$

{ Is it possible this situation on our condition ? }



Respect the two cutting edges



$$h_{D1} = f_z \cdot \cos(\vartheta_1) = f_z \cdot \cos(\varphi_0 / 2)$$

$$h_{D2} = f_z \cdot \cos(-\varphi_0 / 2) = f_z \cdot \cos(\varphi_0 / 2) = h_{D1} = h_D = 0,1 \cdot \cos(36^\circ) = 0,081 \text{ mm}$$

$$\cos(-\alpha) = \cos(\alpha)$$

$\forall f_z$ we have same chip thickness, Symmetry (identical situation)

$$A_{D1} = A_{D2} = A_D = h_D \cdot \overset{5 \text{ mm}}{a_p} = 0,081 \cdot 5 = 0,405 \text{ mm}^2$$

one cutting edge (A) on the left, the other (B) on the right should be inside φ_2 for being a possible situation

because $\varphi_2 = 41,8^\circ > 36^\circ$ OK

total cutting both ϑ_1, ϑ_2

WORST SITUATION (feasible)

$$K_c = K_{c0.4} \left(\frac{0.4}{h_D} \right)^x = 1600 \cdot \left(\frac{0.4}{0.081} \right)^{0.29} = 2542.5 \text{ MPa}$$

definition

$$F_c = F_{c1} = F_{c2} \text{ identical} = 2542.5 \cdot 0.405 = 1029.7 \text{ N}$$

$$T_c = F_{c1} \frac{D/2}{1000} + F_{c2} \frac{D/2}{1000} = F_c \frac{D/2}{1000} \cdot 2 = 1029.7 \frac{60}{1000} = 61.78 \text{ N.m}$$

2 simultaneously working cutting edges Same contrib.

to compute the POWER

WE need to MAKE a choice ! respect the cutting speed

Maximum chip thickness [mm]	0.05	0.1	0.2
Cutting speed [m/min]	305	275	225

being $h_D = f_z \cos(\theta)$

← take a value ?

$$h_{D\text{MAX}} = h_D(\theta=0) =$$

$$= f_z = 0.1$$

being $h_D = 0.1 \text{ mm MAX,}$

$V_c = 275$ of WORST CASE SCENARIO !

Read properly the table ← (DO NOT use an INTERPOLATION of h_D)

$$\Rightarrow \text{given } V_c \rightarrow \omega = \frac{1000}{60} \cdot \frac{V_c}{D/2} = \frac{1000}{60} \cdot \frac{275}{30} = 152.78 \frac{\text{rad}}{\text{s}}$$

$$P_c = T_c \cdot \omega = 61.78 \cdot 152.78 = 9439 \text{ W} \approx 9.4 \text{ kW}$$



Example 4

4

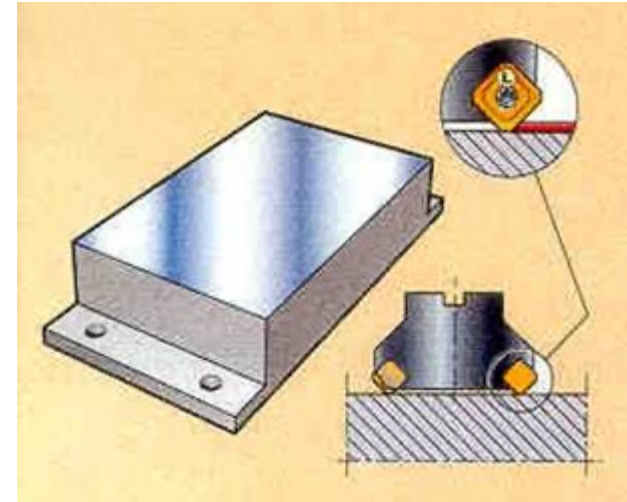
A cast iron plate, with hardness HB 120 and size 100 x 60 mm, needs to be machined by a milling operation. The amount of material allowance to be remove is equal to 8 mm.

Determine:

- a) Chip thickness
- b) Force, torque and power
- c) Milling time

considering both peripheral milling and face milling.

Both situations should consider the following parameters $k_{cs} = 900$ MPa, $x = 0,28$, $e_{in} = e_{out} = 5$ mm, $v_c = 100$ m/min, and $f_z = 0,3$ mm/rev.



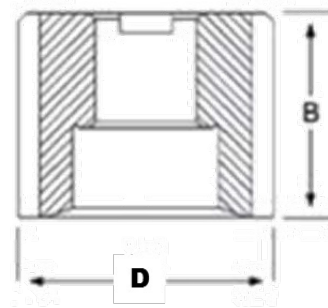


Example 4

5

Slab milling tool:

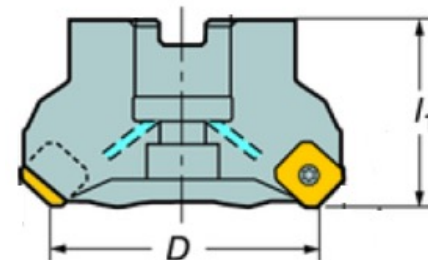
- $B = 80$ mm
- $D = 32$ mm
- $Z = 8$



Face milling tool:

- $l_1 = 60$ mm
- $D = 100$ mm
- $Z = 12, \kappa_r = 45^\circ$
- Square inserts ($l = 9$ mm), $\kappa_r = 45^\circ$
- $D > L$ ($1,3 L < D < 1,4 L$)
- Entrance length $\approx 0,1 D$
- Exit length $\approx 0,3 D$

*bigger diameter tool
(using a square inserts)*



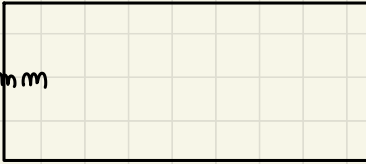
*from
table*

We use THIS

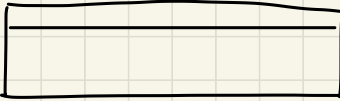
 $l_a = 0,4d$	 $l_a = 2/3 l$	 $l_a = 2/3 l$	 $l_a = 1/2 l$
 $l_a = 1/2 l$	 $l_a = 1/2 l$	 $l_a = 1/4 l$	 $l_a = 1/4 l$

100 mm

60 mm



8 mm



a) h_D

b) F_c, T_c, P_c

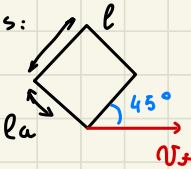
c) t_m ($e_{in} = e_{out} = 5 \text{ mm}$)

$$\begin{cases} K_{cs} = 300 \text{ MPa} \\ \chi = 0,28 \end{cases}$$

$$\begin{cases} V_c = 100 \text{ m/min} \\ f_z = 0,3 \frac{\text{mm}}{\text{REV}} \end{cases}$$

• **FACE MILLING**: $K_r = 45^\circ$ so the tool

Insert is:



$l = 9 \text{ mm}$ (of insert side)

$l_a = 2/3 l = 6 \text{ mm}$ (we don't use all insert length BUT just this portion)

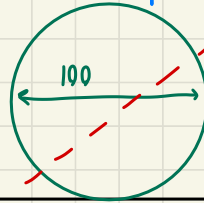
We have info on

the DATA about how to use the tool

from 100 mm tool $\rightarrow$ performing face milling on 60×100 workpiece

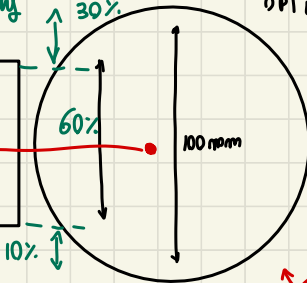
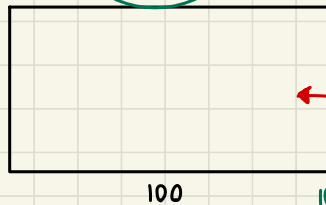
$D = 100 \text{ mm}$
 $Z = 12$

We have 2 choices to position the tool!



OPTION 2 NOT GOOD choice, all diameter involving cutting

OPTION 1

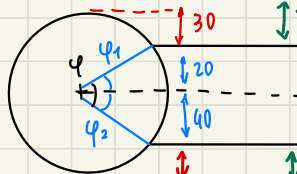


SITUATION OF WORK

using 50% of the tool during cutting considering cutting conditions as told on the data

better to approach from the side! using l_a to cut

$\Rightarrow$ we use all usable portion l_a



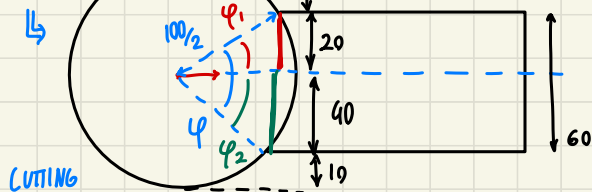
EXIT length (0,3D)

$\leftarrow$ POSITIONING!

as specified by the DATA

$\leftarrow$ ENTRANCE length (0,1D)

TOP VIEW



CUTTING GEOMETRY

ENGAGEMENT ANGLE φ :

Overall situation?

$$\varphi_0 = 2\pi / Z = 2\pi / 12 = 30^\circ \quad (\text{spacing of teeth})$$

$$= 0,524 \text{ rad}$$

$$z = \varphi / \varphi_0 = 76,7 / 30 = 2,56$$

Cutting situation

from this:

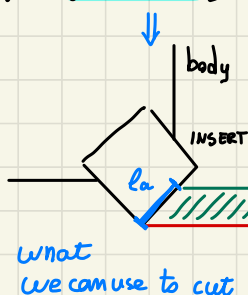
$$\begin{cases} \frac{100}{2} \cdot \sin \varphi_1 = 20 \\ \frac{100}{2} \cdot \sin \varphi_2 = 40 \end{cases} \Rightarrow \begin{cases} \varphi_1 = \arcsin(2/5) = 23,6^\circ \\ \varphi_2 = \arcsin(4/5) = 53,1^\circ \end{cases}$$

$$\varphi = \varphi_1 + \varphi_2 = 76,7^\circ$$

$$= 1,34 \text{ rad}$$

$z > 2$ simultaneously
cutting teeth bigger than 2
↓
AVERAGE APPROACH

there is another issue to check... about Geometry
from {SIDE VIEW}



What is MAX amount of material
that we are able to remove?

a_p : axial depth of cut (max a_p we can remove?)

$$a_{p\text{MAX}} = a_a \cdot \sin(K_r) =$$

$$= 6 \text{ mm} \cdot \sin(45^\circ) = 4,24 \text{ mm}$$

← (max axial depth of cut this tool can remove)
↓ CONSTRAINTS!

because we have to
machine 8mm of depth of cut...

↓ we have to do 2 STEPS! choosing 4mm = a_p ✓ cutting step
2 STEPS of 4mm



2 steps
of cutting

$$\rightarrow a_{p1} = a_{p2} = 4 \text{ mm}$$

(due to CONSTRAINTS of tool)

IMPORTANT aspect to understand the settings!

AVERAGE APPROACH

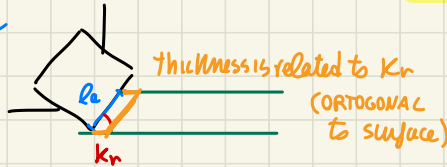
→ we refer to av. thickness $\downarrow$ (from theoretical result...)

$$h_{D,AV} = \frac{2 f_z \cdot a_e}{\varphi \cdot D} \cdot \sin(K_r) = \frac{2 \cdot 0,3 \cdot 60}{1,34 \cdot 100} \cdot \sin(45^\circ) = 0,19 \text{ mm}$$

↑
IN RAD

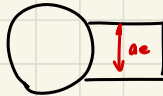
relevant to consider

TOOL USED



a_e : radial depth

of cut is the full diameter $D = 60 \text{ mm}$
all involved



$$A_{D,AV} = h_{D,AV} \cdot a_p / \sin(K_r) =$$

non x length of cutting edged involving cutting because of Geometry

$$= 0,19 \cdot \frac{4}{\sin(45)} = 1,075 \text{ mm}^2$$

evaluated for it

$$K_{C,AV} = K_C(h_{D,AV}) = \frac{K_{CS}}{(h_{D,AV})^{0,28}} = \frac{900}{(0,19)^{0,28}} = 1432,8 \text{ MPa}$$

from here we can compute

$$F_{C,AV} = K_{C,AV} \cdot A_{D,AV} = 1432,8 \cdot 1,075 = 1540 \text{ N}$$

$$T_C = 2 F_{C,AV} \cdot \frac{D/2}{1000} = 2,56 \cdot 1540 \cdot \frac{100/2}{1000} = 197 \text{ Nm}$$

we summm together all

2 CONTRIBUTION (TOTAL TORQUE)

to compute P_c we need

$$V_c = 100 \text{ mm/min} \rightarrow n = 1000 \cdot \frac{100}{\pi \cdot D} = 1000 \cdot \frac{100}{\pi \cdot 100} = 318 \text{ rpm}$$

$$\omega = \frac{2\pi n}{60} = \frac{2\pi}{60} \cdot 318 = 33,3 \frac{\text{rad}}{\text{s}}$$

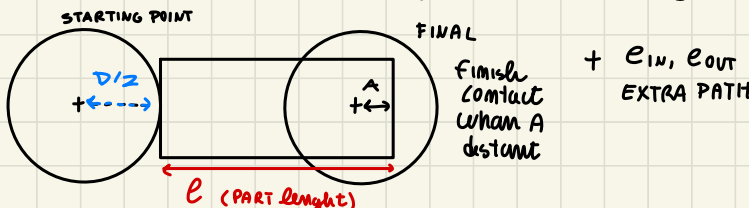
Small rev per min!

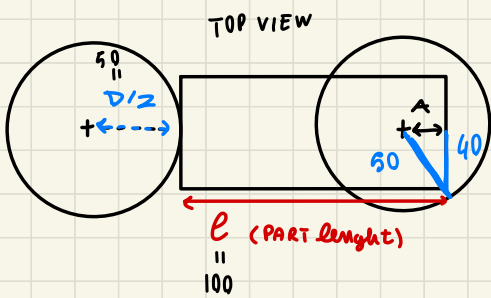
Requirements

$$P_c = T_C \cdot \omega = 197 \cdot 33,3 = 6560 \text{ W} = 6,6 \text{ kW}$$

identical requirements of each step, so same requirements

CUTTING TIME: we can evaluate from the situation of work





$$A = \sqrt{50^2 - 40^2} = 30 \text{ mm}$$

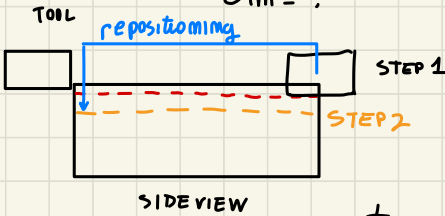
$$t_{m1} = \frac{e + D/2 - A + e_{in} + e_{out}}{12 \cdot 0,3 \cdot 318} \cdot [60] = 6,8 \text{ s}$$

STEP 1

$f = (Z \cdot f_z \cdot m)$

TO PERFORM the overall operation (2 STEPS)

$$t_m = ?$$



to perform step 2 we have to
reposition the tool (perform 2 on same
cutting condition)
and to the same...

$$t_{m1} + t_{m2} + \bar{t}_R$$

t_{m1}
same
condition

rapid time to go back
and start second step
(REPOSITIONING)

$$t_m = 2 \times 6,8 + t_R$$

$$= 13,6 + t_R \text{ [s]}$$

($t_m > 13,6$ to have an idea about overall machining time)

We compute all requirements about **FACE MILLING** (OPTION 1)

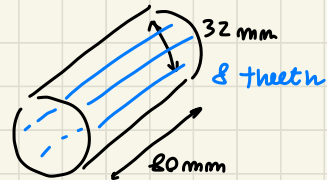
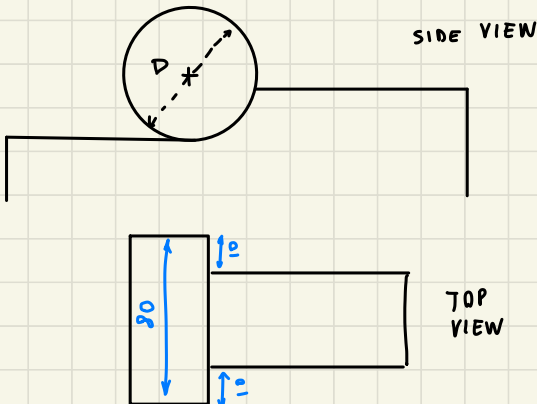
BUT we have a

SECOND OPTION with same process parameter but different tool
SLAB MILLING different cutting geometry and WORKING condition
(due to different tool)

$$D = 32 \text{ mm}$$

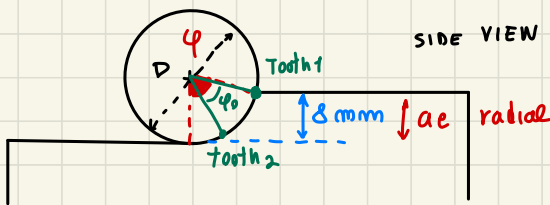
$$Z = 8$$

$$B = \text{length of the tool} = 80 \text{ mm}$$



We can involve the
tool for 60 mm
on the workpiece leaving
10 mm on the two sides

to answer we must understand
CUTTING GEOMETRY



We suppose to cut all 8 in one row

$$a_e = 8 \text{ mm}$$

$$a_p = 60 \text{ mm}$$

$$a_e = \frac{D}{2} - \frac{D}{2} \cos(\varphi)$$

$$\cos \varphi = \frac{D/2 - a_e}{D/2} = 1 - \frac{2a_e}{D}$$

$$\varphi = \arccos\left(1 - \frac{2a_e}{D}\right) = \arccos\left(1 - \frac{2 \cdot 8}{32}\right) = 60^\circ = 1,05 \text{ rad}$$

$$\varphi_0 = 2\pi/Z = 2\pi/8 = 0,785 \text{ rad}$$

$$Z = \varphi/\varphi_0 = 1,05/0,785 = 1,34$$

$Z \in (1,2) \rightarrow$ NOT AV. approach
WORST CASE!

Worst case solution

(TOOTH 1) in position $\vartheta_1 = \varphi = 60^\circ$

$$h_{D1} = h_{D\text{MAX}} = 0,3 \cdot \sin(\vartheta_1) = 0,26 \text{ mm}$$

$$A_{D1} = A_{D\text{MAX}} = h_{D\text{MAX}} \cdot a_p = 0,26 \cdot 60 = 15,6 \text{ mm}^2$$

$$K_{c1} = \frac{K_{cs}}{(h_{D1})^x} = \frac{300}{(0,26)^{0,28}} = 1312 \text{ MPa}$$

$$F_{c1} = K_{c1} \cdot A_{D1} = 1312 \cdot 15,6 = 20467 \text{ N} = 20,5 \text{ kN}$$

IF tooth 1 is @ $\vartheta_1 = \varphi$, tooth 2 is on $\vartheta_2 = \varphi - \varphi_0$ ($\varphi_0 = 45^\circ$)

(TOOTH 2) in position $\vartheta_2 = \varphi - \varphi_0 = 60 - 45 = 15^\circ$

$$h_{D2} = 0,3 \sin(15) = 0,078 \text{ mm}$$

$$A_{D2} = 0,078 \cdot 60 = 4,7 \text{ mm}^2$$

$$K_{c2} = \frac{300}{(0,078)^{0,28}} = 1838 \text{ MPa}$$

$$F_{c2} = 1838 \cdot 4,7 = 8639 \text{ N} = 8,6 \text{ kN}$$

TWO
CONTRIBUTION
TO THE TORQUE
TOTAL
 $\hookrightarrow$
IN THIS
WORST
CASE

$$T_c = 20467 \cdot \frac{32/2}{1000} + 8639 \cdot \frac{32/2}{1000} = 327,5 + 138,2 = 465,7 \text{ Nm}$$

TOTAL TORQUE

different tool D

FIRST CUTTING edge has more contrib.

$$V_c = 100 \text{ m/min} \rightarrow n = \frac{1000}{\pi \cdot 32} = 995 \text{ rpm}$$

$$\omega = \frac{2\pi}{60} \cdot 995 = 104,17 \frac{\text{rad}}{\text{s}}$$

$$\rightarrow P_c = T_c \cdot \omega = 465,7 \cdot 104,17 = 48511,969 \text{ W} = 48,5 \text{ kW}$$

SCAB MILLING

composed of ONLY ONE STEP

$P_{c \text{ SLAB}} = 48,5 \text{ kW}$ relevant POWER Requirements

While FACE MILLING of 2 STEP

$P_{c \text{ FACE}} = 6,6 \text{ kW} \ll P_{c \text{ SLAB}}!$

IF we have η pe limits...

IF we have $P_c \geq \eta$ pe $\times$ → you should divide the process into STEPS! (2, 3...)

(using identical steps)

solve less power $\forall$ STEP

NOTICE

→ IF you solve using

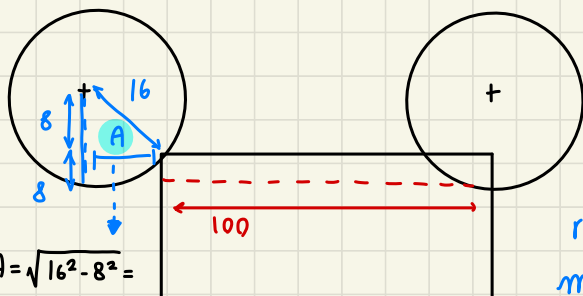
AVERAGE APPROACH

~~$P_c = 28,6 \text{ kW}$~~

WRONG totally underestimating the REQUIREMENTS!

for the

CUTTING TIME



t_m cut

$$t_m = \frac{100 + 13,9 + 5 + 5}{8 \cdot 0,3 \cdot 995} [60] = 3,1 \text{ s}$$

$$A = \sqrt{16^2 - 8^2} = 13,9 \text{ mm}$$

removing huge amount of material $\times$ unit of time requiring relevant P_c , so with small t_m

MAXIMIZE MRR

While on face milling

We require more t_m ...

BEST SOLUTION SLAB MILLING

if you have enough η pe

on Roughing the goal is the removal fast considering the constraints (NOT the precision of finish) (comparing with same process parameter the two options)



Example 5

6

A steel square planar surface with a length equal to 300 mm must be flattened. A roughing operation will remove a machining allowance equal to 4 mm. The face mill will be positioned initially so that 25% of the tool diameter will exit from one side of the plane width and the square plane is machined according to the path in the figure.

↳ next page

Determine:

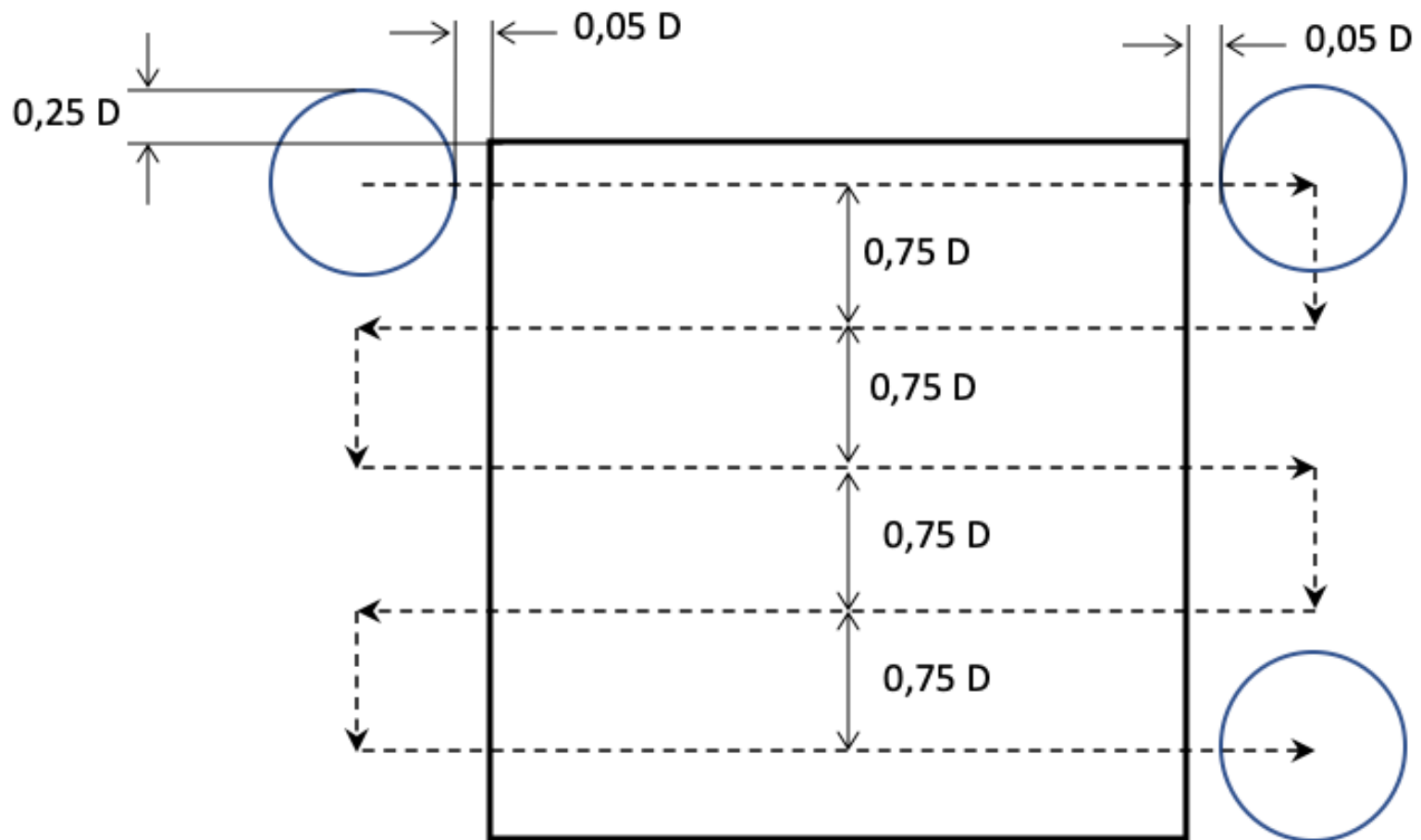
- 1) The cutting torque and the cutting power, verifying that the operation can be carried out with a milling machine with a power of 15 kW and an efficiency of 80%.
- 2) The machining time needed to carry out the operation.

Data: mill diameter $D = 80$ mm; number of mill cutting edges $Z = 8$; main entering angle equal to 90° ; Kronenberg parameters $k_{cs} = 1600$ MPa, $x = 0,197$; feed per tooth $f_z = 0,15$ mm/rev; cutting speed $v_c = 250$ m/min.

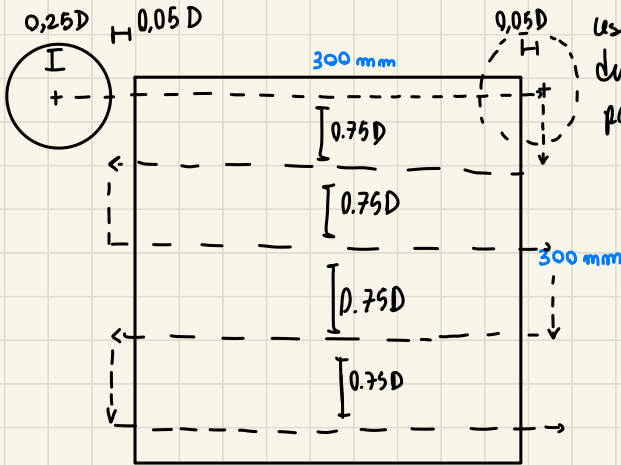


Example 5

7



TOP VIEW



We are flattening the surface using a smaller tool than plane dimension, moving the tool parallel to previous step

we can use 0,75D of the tool VSTEP (use 75% of tool diameter) doing 5 PARALLEL steps to remove all Material!

DATA

$$D = 80 \text{ mm}$$

$$Z = 8 \quad K_r = 90^\circ$$

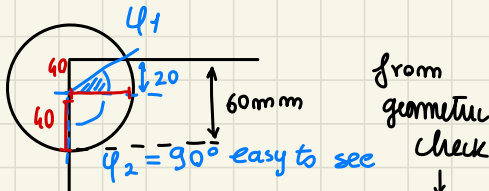
$$\text{KRONENBERG} \left\{ \begin{array}{l} K_{cs} = 1600 \text{ MPa} \\ \chi = 0.196 \end{array} \right.$$

$$f_z = 0,15 \text{ mm/rev}$$

$$V_c = 250 \text{ m/min}$$

$$a_p = 4 \text{ mm} \quad (\text{what we remove})$$

$$a_e = 0,75 \cdot 80 = 60 \text{ mm} \quad (\text{used part of the tool})$$



$$\phi_1 = \arcsin\left(\frac{20}{40}\right) = 30^\circ$$

$$\phi = \phi_1 + \phi_2 = 30 + 90 = 120^\circ = 2,1 \text{ Rad}$$

$$\phi_0 = 2\pi/Z = 2\pi/8 = 45^\circ$$

$$z = \phi/\phi_0 = 120/45 = 2,67$$

$z > 2$ we can use Average approach!

$$h_{D,AV} = \frac{f_z}{\phi} \cdot \frac{2a_e}{D} = \frac{0,15 \cdot 2 \cdot 60}{2,1 \cdot 80} = 0,107 \text{ mm}$$

$$(\sin(K_r) = 1)$$

$$A_{D,AV} = h_{D,AV} \cdot a_p = 0,107 \cdot 4 = 0,428 \text{ mm}^2$$

amount of remove material

$$K_{C,AV} = \frac{1600}{(0,107)^{0,197}} = 2485 \text{ MPa}$$

$$F_{C,AV} = 2485 \cdot 0,428 = 1064 \text{ N}$$

$$T_C = 2,67 \cdot \frac{1064}{2} \cdot \frac{80/2}{1000} = 113,64 \text{ N m}$$

$$\omega = \frac{1000}{60} \cdot \frac{250}{80/2} = 104,2 \text{ rad/s}$$

$$P_c = T_c \cdot \omega = 113,64 \cdot 104,2 = 11841 = 11,8 \text{ kW}$$

check for feasibility:

$$P_H = P_c \eta = 0,8 \cdot 15 = 12 \text{ kW} > 11,8 \text{ kW} \quad \text{OK Feasible operation}$$

MACHINE
POWER

TIME

to compute time we need $n = \frac{1000 \cdot 250}{\pi \cdot 80} = 995 \text{ rpm}$

$$V_f = 8 \cdot f_z \cdot n = 8 \cdot 0,15 \cdot 995 = 1194 \text{ mm/min}$$

as we can see, feed velocity $\ll V_c$

we neglect feed contribution
to POWER P_c because negligible
 $V_f \ll V_c$ REMEMBER

↓

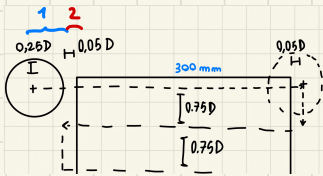
We should compute TOTAL path

length:

$$l_m = \left[\left(\frac{D}{2} + 0,05D \right) 2 + \left(l_p \right) 5 + \left(0,75D \right) 4 \right] = \left[(1,1) \cdot 80 + 300 \right] 5 + 0,75 \cdot 80 \cdot 4 = 2180 \text{ mm}$$

1 2 @ beginning and end vertical motion to reposition
 PART LENGTH reposition motion 4 total reposition
 do me 5 times

TOTAL MACHINING LENGTH



$$t_m = \frac{l_m}{V_f} = \frac{2180}{1194} = 1,83 \text{ min} = 110 \text{ s}$$

being a small motion here we maintain same V_f , of different steps
with different condition is a different evaluation